

[illegible]

**B.Tech/ M.Tech (Integrated) DEGREE EXAMINATION, MAY 2024**

### Sixth Semester

21CSS303T – DATA SCIENCE

*(For the candidates admitted from the academic year 2022-2023 onwards)*

**Note:**

- (i) **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40<sup>th</sup> minute.
- (ii) **Part - B** and **Part - C** should be answered in answer booklet.

Time: 3 Hours

Max. Marks: 75

**PART – A (20 × 1 = 20Marks)**

**Answer ALL Questions**

| PART – A (20 × 1 = 20Marks)                                                                                                                                                                                                                                         |   | Marks | BL | CO | PO |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-------|----|----|----|
| Answer ALL Questions                                                                                                                                                                                                                                                |   |       |    |    |    |
| 1. Which of the following is used to reshape a NumPy array?<br>(A) reshape () (B) resize ()<br>(C) Both (A) and (B) (D) size ()                                                                                                                                     | 1 | 2     | 1  | 1  |    |
| 2. A column is a _____ representation of data.<br>(A) Diagonal (B) Vertical<br>(C) Top (D) Horizontal                                                                                                                                                               | 1 | 1     | 1  | 1  |    |
| 3. Analyze and choose the correct output of the following code?<br>import numpy as np<br>a = np.array ([1,2,3])<br>b = np.array ([4,5,6])<br>c = np.concatenate ((a,b)) print c<br>(A) [[1,2,3,], [4,5,6]] (B) [[1,4], [2,5], [3,6]]<br>(C) [1,2,3,4,5,6] (D) Error | 1 | 4     | 1  | 1  |    |
| 4. Identify which of the following is used to analyze the data in Pandas.<br>(A) Dataframe (B) Series<br>(C) Both (A) and (B) (D) Frame                                                                                                                             | 1 | 2     | 1  | 1  |    |
| 5. Which of the following is correct to access individual item from dataframe df.<br>(A) df.iat [2, 2] (B) df.loc [2, 2]<br>(C) df.at [2, 2] (D) df [0, 0]                                                                                                          | 1 | 2     | 1  | 1  |    |
| 6. What will be output of following code?<br>import numpy as np<br>array1 = np.array ([100, 200, 300, 400, 500, 600, 700])<br>print (array1 [1:5:2])<br>(A) [200 300] (B) [200 700]<br>(C) [200 400] (D) [200 6300]                                                 | 1 | 3     | 1  | 1  |    |
| 7. What does the 'eye( )' function in NumPy do?<br>(A) Generates an identity matrix (B) Performs element-wise multiplication<br>(C) Calculates the dot product of two arrays (D) Splits an array into sub_arrays                                                    | 1 | 1     | 1  | 1  |    |

8. Point out the correct statement. 1 1 2 1
- (A) If data is a list, if index is passed the values in data corresponding to labels in index will be pulled out (B) NaN is the standard missing data marker used in Pandas
- (C) Series acts very similarly to a array (D) Dataframe acts similar to series
9. To extract first three rows and three columns of a dataframe, 'exp' find the true statement. 1 2 2 1
- (A) exp.iloc [0:2, 0:2] (B) exp.iloc [0:3, 0:3]
- (C) exp.iloc [1:3, 1:3] (D) exp.iloc [1:4, 1:4]
10. What are some challenges associated with handling a large volume of data? 1 2 2 1
- (A) Limited processing power and insufficient storage (B) Slow algorithms and network bandwidth limitation
- (C) Overloaded memory and algorithms running indefinitely (D) Lack of proper data visualization techniques and GPU underutilization
11. Recognize which resource limitation can lead to speed issues in data analysis. 1 3 2 1
- (A) Limited RAM capacity (B) Input / output (I/O) and CPU starvation
- (C) Insufficient network bandwidth (D) Lack of proper algorithms
12. Identify the issue that may arise when processing large datasets. 1 2 2 1
- (A) Limited GPU performance (B) Incompatibility with modern CPU architectures
- (C) Bottlenecks in computer components (D) Lack of proper data visualization techniques
13. What is one solution mentioned for addressing slow data processing due to reading data from hard-drives? 1 1 2 1
- (A) Increasing RAM capacity (B) Upgrading to a faster CPU
- (C) Using solid state drives instead of hard disk drives (D) Optimizing algorithm efficiency
14. Identify which of the following is a common approach to identify and handle outliers or anomalies in a dataset? 1 1 2 1
- (A) Replacing them with mean values (B) Ignoring them during analysis
- (C) Detecting and removing them (D) Treating them as missing data
15. What solution category is emphasized as potentially solving more problems than adding more or better hardware? 1 1 2 1
- (A) Increasing CPU speed (B) Using more durable hard disk drives
- (C) Choosing the right algorithms (D) Upgrading RAM capacity

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|-------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|
| 16. What parameter of the <code>plt.subplots()</code> function is used to specify number of rows and columns in subplot grid? | 1 | 1 | 4 | 1 |
| (A) rows                                                                                                                      |   |   |   |   |
| (B) columns                                                                                                                   |   |   |   |   |
| (C) nrow                                                                                                                      |   |   |   |   |
| (D) ncols                                                                                                                     |   |   |   |   |
|                                                                                                                               |   |   |   |   |
| 17. Which matplotlib function is used to set the title of a plot?                                                             | 1 | 1 | 4 | 1 |
| (A) <code>plot.title()</code>                                                                                                 |   |   |   |   |
| (B) <code>plt.set_title()</code>                                                                                              |   |   |   |   |
| (C) <code>plt.title()</code>                                                                                                  |   |   |   |   |
| (D) <code>plot.set_title()</code>                                                                                             |   |   |   |   |
|                                                                                                                               |   |   |   |   |
| 18. In seaborn, which function is used to create a scatterplot with optional hue and style semantics?                         | 1 | 1 | 4 | 1 |
| (A) <code>seaborn.scatterplot()</code>                                                                                        |   |   |   |   |
| (B) <code>seaborn.scatter()</code>                                                                                            |   |   |   |   |
| (C) <code>seaborn.plot.scatter()</code>                                                                                       |   |   |   |   |
| (D) <code>seaborn.scatter_plot()</code>                                                                                       |   |   |   |   |
|                                                                                                                               |   |   |   |   |
| 19. Which parameter of the <code>plt.hist()</code> function in matplotlib controls the number of bins in a histogram?         | 1 | 1 | 5 | 1 |
| (A) bins                                                                                                                      |   |   |   |   |
| (B) bar                                                                                                                       |   |   |   |   |
| (C) density                                                                                                                   |   |   |   |   |
| (D) orientation                                                                                                               |   |   |   |   |
|                                                                                                                               |   |   |   |   |
| 20. Which seaborn function is used to create a box plot?                                                                      | 1 | 1 | 4 | 1 |
| (A) <code>seaborn.boxplot()</code>                                                                                            |   |   |   |   |
| (B) <code>seaborn.plot_box()</code>                                                                                           |   |   |   |   |
| (C) <code>seaborn.plot.boxplot()</code>                                                                                       |   |   |   |   |
| (D) <code>seaborn.box_plot()</code>                                                                                           |   |   |   |   |

**PART – B (4 × 10 = 40 Marks)**

Answer ANY FOUR Questions

- |                                                                                                                                                                                                                                                         | Marks | BL | CO | PO |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|----|----|----|
| 21. Illustrate the lifecycle of data science process with real time application.                                                                                                                                                                        | 10    | 2  | 1  | 1  |
| 22. Discuss the role of APIS in web scraping. How do APIS facilitate data extraction from websites and what are the advantages of using them compared to traditional scraping methods? Provide examples of popular APIS commonly used for web scraping. | 10    | 2  | 2  | 1  |
| 23. List the common challenges associated with managing large datasets and equate how these challenges can influence the effectiveness of data analysis and processing tasks.                                                                           | 10    | 3  | 3  | 1  |
| 24. Explain string manipulation with python code example.                                                                                                                                                                                               | 10    | 2  | 4  | 1  |
| 25. Elaborate the common formatting options available for ticks, labels and legends in matplotlib and provide examples of when and how to use them effectively.                                                                                         | 10    | 3  | 5  | 1  |
| 26. Write python program to plot histogram by assuming your own data and explain the various attributes of histogram.                                                                                                                                   | 10    | 3  | 5  | 1  |

**PART – C (1 × 15 = 15 Marks)**

Answer ANY ONE Question

- |                                                                        | Marks | BL | CO | PO |
|------------------------------------------------------------------------|-------|----|----|----|
| 27. Evaluate the facet of data and illustrate with real time examples. | 15    | 4  | 1  | 1  |

- 28.i. List some common types of plots that can be created using Pandas and seaborn explain how can they be generated using Python code. 8 3 4 1
- ii. Explain how can python be utilized for data summarization and binning tasks. 7 3 5 1  
What are the key methods involved in performing these operations?

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